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L2

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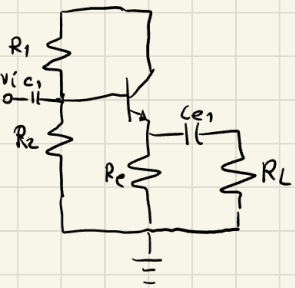
Etapa 3

Tip 122, $V_{cc} = 24V$, $\beta = 3000$, $I_{CQ} = 700mA$

$V_{cc} = V_C$, $V_{ce} \cong 12V$, $R_L = 8\Omega$, $A_v \approx 1$

$$r_e = \frac{26mV}{I_{CQ}}$$



$V_e = V_C - V_{ce}$

$V_e = 12V$

$$R_e = \frac{V_e}{I_e}$$

$R_e = 17\Omega \approx 20\Omega$

$r_e = 37.15m\Omega$

$P_{Re} = I_{CQ}^2 R_e$

$P_{Re} = 9.8W$

Al menos > 10W

$C_1 = 10\mu F$

$V_B = V_e + V_{Be}$

$V_B = 13.4V$

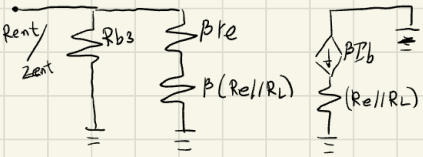
$$R_b \leq \frac{1}{10}\beta R_e$$

$$R_1 = \frac{V_{cc} R_b}{V_B} = 9134 \approx 9.2K\Omega$$

$R_b \leq 5.703K\Omega$

$$R_2 = \frac{R_b}{1 - \frac{V_B}{V_{cc}}} = 11.5K \approx 15K\Omega$$

Tomando el modelo de parámetros re



$Z_{ent} = R_{b3} // \beta(r_e + R_e//R_L) \rightarrow \{\beta(r_e + R_e//R_L) = 5.76\Omega$

$Z_{ent} = 5.703K // (3000)(5.76\Omega)$

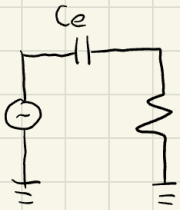
$Z_{ent} \cong 4.29K\Omega$

$$A_{v3} = \frac{R_e//R_L}{r_e + R_e//R_L} = \frac{5,7}{37.15m + 5,7}$$

$A_{v3} = 0.993$

$$A_{I3} = A_{v3}\left(\frac{R_{en3}}{R_L}\right)$$

$A_{I3} = 532$



C mínimo (f = 20Hz)

$X_C < R_L$

$$\frac{1}{2\pi f_C} < R_L$$

$$C > \frac{1}{2\pi f R_L}$$

$C > 9000\mu F$

$C = 10.000\mu F$

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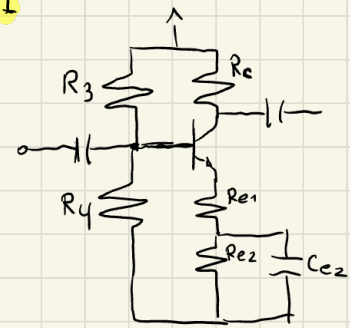
- Etapa 2 → Emisor común

$r_e = 13\Omega$



$R_{L3} = 4{,}29K\Omega$, $I_{CQ} = 2mA$, $\beta = 250$

$A_{v_2} = -26$, $V_{CEQ} = 12V$

$R_e = \frac{V_e}{I_e} = 1.2K\Omega$

$I_b = \frac{I_c}{\beta} = 8\mu A$

$R_e = R_{e1} + R_{e2}$

$R_b \leq \frac{1}{10}\beta R_e$

$R_b \leq 30K\Omega$

Condición de diseño

$V_e = 10 - 15\% V_{cc}$

$V_e = 10\% V_{cc} = 2.4V$

$A_{v2} = \frac{R_c // R_{L3}}{r_e + R_{e1}} \Rightarrow A_{v2}(r_e + R_{e1}) = 2243 \rightarrow r_e + R_{e1} = \frac{2243}{-26}$

$R_{e1} \simeq 86{,}3 - 13 = 73\Omega \approx 75\Omega$

$V_B = V_E + V_{BE}$

$V_B = 3.1V$

$R_e - R_{e1} = R_{e2} \rightarrow R_{e2} = 1.1K\Omega$

$R_{e1} = 100\Omega$ → Valor comercial

↳ Lo pusimos en 1K en la simulación y funciona también, pero se decidió poner ese valor solo para la resistencia en físico

$I_{div} = 10 I_b = 80\mu A \rightarrow R_1 = \frac{V_{cc} - V_B}{I_{div}} = 261K \approx 270K\Omega$

$R_1 = \frac{V_{cc} R_b}{V_B} = 232K$

$R_1 \approx 270K\Omega$ → V. Comercial

$V_{RC} = V_{cc} - V_c - V_{ce}$

$V_{RC} = 9.6V$

$R_2 = \frac{V_B}{I_{div}} = 38.7K \approx 39K\Omega$

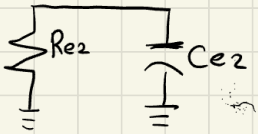
$R_2 = \frac{R_b}{1 - \frac{V_B}{V_{cc}}} = 34{,}5K$

$R_2 \approx 39K\Omega$ → V. Comercial

$R_c = \frac{V_{RC}}{I_c} = 4.8K\Omega \approx 4.7K\Omega$

$x_c < R_{e2}$

$\frac{1}{2\pi f C} < R_{e2}$



$C_{e_2} > \frac{1}{2\pi f R_{e2}} \rightarrow 7{,}23\mu F$

$C_{e2} = 10\mu F$

↓

Este valor lo hemos seleccionado y variado con otros valores, pero para que la señal no se distorsione lo bajamos a 2.2μF, pero ese valor también sirve.

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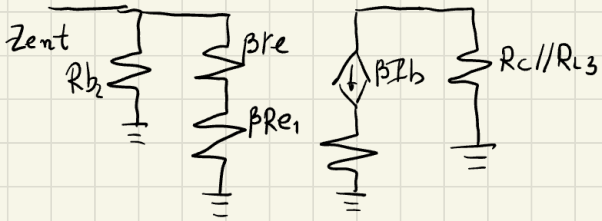

$$A_{I2} = A_{V2}\left(\frac{R_{en2}}{R_{en3}}\right)$$

$$A_{I2} = 87{,}8$$

$Zent_2 = Rb_2 // \beta(re+Re1) \rightarrow \beta(re+Re1) = 28250$

$Zent_2 = 30K // 28250$

$Zent_2 = 14{,}5K\Omega$

- Etapa 1 bc547B, $\beta = 250$, $I_{CQ} = 2mA$, $V_{CEQ} = 12V$, $R_{L2} = 14.5K\Omega$

$A_{V1} = -21$

* Para esta etapa, casi todos los valores son los mismos que la etapa 2 menos: A_{V1}, Re_1 y Re_2 y R_C

$V_{RC} = V_{CC} - V_C - V_{CE}$

$V_{RC} = 9.6V$

$R_1 = 270K$

$R_2 = 39K$

$V_e = 10\% V_{CC}$

$V_e = 2.4V$

$V_B = 3.1V$

$V_C = V_e + V_{CE}$

$V_C = 2.4V$

$Re = 1.2K\Omega \rightarrow Re_1 = 180$, $Re_2 = 1K\Omega$

$R_C = \frac{V_{RC}}{I_C} \cong 4.7K \rightarrow$ Aunque da ese valor, la ganancia no daría -21, sino igual a la etapa 2, asique probando valores, el valor adecuado para ese resultado, para RC sería 5.7K Ω

$$A_{V1} = -\frac{R_C // R_{L2}}{re + Re_1} = -\frac{4091}{13+180} = -21{,}1 //$$

$$A_{I1} = A_{V1}\left(\frac{R_{en}}{R_{en2}}\right) \approx 26{,}92$$

$Rb \leq 0.1\beta Re$

$Rb \leq 30K\Omega$

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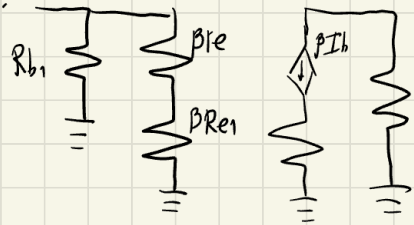
Zent1 :



$Zent_1 = Rb_1 // \beta(re+Re_1) \longrightarrow \{\beta(re+Re_1) = 48250$

$Zent_1 = 30K // 48250$

$Zent_1 = 18{,}5K\Omega$

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